Mat2002 Tutorial 12

Junwei Zha

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Laplace transformaiton

Suppose that $f(t)$ is a real function defined on $[0, +\infty)]$ and
(1) f is piecewise continuous on the interval $0 \leq t \leq A$ for any positive A.
(2) $|f(t)| \leq Ke^{at}$ when $t \geq M$. In this inequality, K, a, and M are real constants $(K > 0, M > 0)$.($f(t)$ is in the "exponential" oreder.)

Then the Laplace Transform $\mathcal{L}f(t)(s) = F(s)$ defined as

$$\mathcal{L}\{f(t)\}(s) = F(s) = \int_0^\infty e^{-st} f(t)dt$$

exists for $s > a$.

Laplace transformaiton

If the Laplace transform $\mathcal{L}$ is defined as

$$\mathcal{L}\{f(t)\}(s) = F(s) = \int_0^\infty e^{-st} f(t)dt$$

then the inverse Laplace transform for $F(s) = \mathcal{L}f(t)$ is defined as

$$\mathcal{L}^{-1}\{F(s)\} = f(t).$$

Thus,

$$\mathcal{L}^{-1}\{F(s)\} = f(t) \Leftrightarrow \mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)dt.$$

Table of Laplace transformation

Table for Elementary Laplace Transforms

	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
1.	1	$\frac{1}{s}, s > 0$
2.	e^{at}	$\frac{1}{s-a}, s > a$
3.	$t^n, n > 0$ integer	$\frac{n!}{s^{n+1}, s>0}$
4.	$t^p, p > -1$	$\frac{\Gamma(p+1)}{s^{p+1}}, s > 0$
5.	$\sin at$	$\frac{a}{s^2+a^2}, s > 0$
6.	$\cos at$	$\frac{s}{s^2+a^2}, s > 0$
7.	$\sinh at$	$\frac{a}{s^2-a^2}, s > \lvert a \rvert$
8.	$\cosh at$	$\frac{s}{s^2-a^2}, s > \lvert a \rvert$

Table of Laplace transformation

	$f(t) = \mathcal{L}^{-1}\{F(s)\}$	$F(s) = \mathcal{L}\{f(t)\}$
9.	$e^{at} \sin bt$	$\frac{b}{(s-a)^2+b^2}, s > a$
10.	$e^{at} \cos bt$	$\frac{s-a}{(s-a)^2+b^2}, s > a$
11.	$t^n e^{at}, n > 0$ integer	$\frac{n!}{(s-a)^{n+1}}, s > a$
12.	$u_c(t)$	$\frac{e^{-cs}}{s}, s > 0$
13.	$u_c(t)f(t-c)$	$e^{-cs}F(s)$
14.	$e^{ct}f(t)$	$F(s-c)$
15.	$f(ct)$	$\frac{1}{c}F\left(\frac{s}{c}\right), c > 0$
16.	$\int_0^t f(t-\tau)g(\tau)d\tau$	$F(s)G(s)$
17.	$\delta(t-c)$	e^{-cs}

Property of Laplace Transform

Theorem

If $\mathcal{L}\{f_1(t)\}(s)$ *and* $\mathcal{L}\{f_2(t)\}(s)$ *are defined for* $s > a$, *and* c_1 *and* c_2 *are any numbers, then* $\mathcal{L}\{c_1 f_1(t) + c_2 f_2(t)\} = c_1\mathcal{L}\{f_1(t)\} + c_2\mathcal{L}\{f_2(t)\}$.

Theorem

Suppose that f *is continuous and* f' *is piecewise continuous on any interval* $0 \leq t \leq A$. *Suppose further that there exist constants* K, a, *and* M *such that* $|f(t)| \leq Ke^{at}$ *for* $t \geq M$. *Then* $\mathcal{L}\{f'(t)\}$ *exists for* $s > a$, *and moreover,*

$$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0).$$

Property of Laplace Transform

Corollary

Suppose that $f, f', \ldots, f^{(n-1)}$ *are continuous and* $f(n)$ *is piecewise continuous on any interval* $0 \leq t \leq A$. *Suppose further that there exist constants* K, a, *and* M *such that* $\left|f^{(i)}(t)\right| \leq Ke^{at}, i = 0, 1, \ldots, n-1$ *for* $t \geq M$. *Then* $\mathcal{L}\left\{f^{(n)}(t)\right\}$ *exists for* $s > a$, *and moreover,*

$$\mathcal{L}\left\{f^{(n)}(t)\right\} = s^n F(s) - s^{n-1} f(0) - \cdots - f^{(n-1)}(0).$$

where $F(s) = \mathcal{L}\{f(t)\}$.

Property of Laplace Transform

If $F(s) = \mathcal{L}\{f(t)\}$ exists for $s > a \geq 0$, and if c is a constant, then

$$\mathcal{L}\left\{e^{ct} f(t)\right\} = F(s-c), \quad s > a + c.$$

Conversely, if $f(t) = \mathcal{L}^{-1}\{F(s)\}$, then

$$e^{ct} f(t) = \mathcal{L}^{-1}\{F(s-c)\}.$$

Solve IVP

The method by using the Laplace transform to solve a differential equation:
(1) Apply the Laplace Transform to the whole equation,
(2) Substitute any intial conditions f_0 and $f'(0)$,
(3) Rearrange and solve for $F(s)$,
(4) Apply the inverse Laplace Transform to find $f(t)$.

Exercise

Use the Laplace Transform to solve the following Second-order Linear ODE:

$$\frac{d^2f}{dt^2} - \frac{df}{dt} - 6f(t) = 0$$

where $f(0) = 3$ and $f'(0) = 4$.

Solution

$$\begin{aligned}
\mathcal{L}\left\{f''(t)\right\} &= s^2F(s) - sf(0) - f'(0), \\
\mathcal{L}\left\{f'(t)\right\} &= sF(s) - f(0), \\
0 &= \mathcal{L}\left\{f''(t)\right\} - \mathcal{L}\left\{f'(t)\right\} - 6\mathcal{L}\{f(t)\}, \\
0 &= s^2F(s) - sf(0) - f'(0) - (sF(s) - f(0)) - 6F(s), \\
F(s) &= \frac{3s+1}{s^2 - s - 6} \\
f(t) &= 2e^{3t} + e^{-2t}
\end{aligned}$$

Step functions & Shifting Theorems

The unit step function or Heaviside function is defined by

$$u_c(t) = \begin{cases} 0, & t < c, \\ 1, & t \geq c, \end{cases} \qquad c \geq 0$$

For a given function f defined for $t \geq 0$, we will often want to consider the related function g defined by

$$y = g(t) = \begin{cases} 0, & t < c, \\ f(t-c), & t \geq c, \end{cases} \qquad = u_c(t) f(t-c)$$

which represents a translation of f a distance c in the positive t direction.

Step functions & Shifting Theorems

Theorem

If $F(s) = \mathcal{L}\{f(t)\}$ *exists for* $s > a \geq 0$, *and if* c *is a positive constant, then*

$$\mathcal{L}\{u_c(t)f(t-c)\} = e^{-cs}\mathcal{L}\{f(t)\} = e^{-cs}F(s), \quad s > a.$$

Conversely, if $f(t) = \mathcal{L}^{-1}\{F(s)\}$, *then*

$$u_c(t)f(t-c) = \mathcal{L}^{-1}\left\{e^{-cs}F(s)\right\}.$$

Exercise

Calculate the Laplace Transform of the following function:

$$f(t) = u_0(t)\sin(t) - u_\pi(t)\sin(t).$$

Solution

$$\begin{aligned}
\sin(t) &= -\sin(t-\pi) \\
f(t) &= u_0(t)\sin(t) + u_\pi(t)\sin(t-\pi) \\
\mathcal{L}\{f(t)\} &= \mathcal{L}\{\sin(t)\} + e^{-\pi t}\mathcal{L}\{\sin(t)\} \\
F(s) &= \frac{1}{s^2+1} + \frac{e^{-\pi t}}{s^2+1}
\end{aligned}$$

Unit impulses and the Dirac delta functions

The delta function has the following properties:

$$\delta(t) = 0, t \neq 0, \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t)dt = 1$$

A unit impulse at an arbitrary point $t = t_0$ is given by $\delta(t - t_0)$, which satisfies

$$\delta(t - t_0) = 0, t \neq t_0, \quad \text{and} \quad \int_{-\infty}^{\infty} \delta(t - t_0)\, dt = 1.$$

Unit impulses and the Dirac delta functions

Theorem

Let $f : \mathbb{R} \to \mathbb{R}$ be a integrable function and continuous near t_0 then

$$\mathcal{L}\{\delta(t - t_0) f(t)\} = e^{-t_0 s} f(t_0)$$

Exercise

Use the Laplace Transform to solve the following Second-order Linear ODE:

$$\frac{d^2f}{dt^2} + 2\frac{df}{dt} - 15f(t) = 8\delta(t-9)$$

where $f(0) = -5$ and $f'(0) = 7$.

Solution

By taking the Laplace transform of the differential equation and using the initial conditions, we have

$$\left(s^2+2s-15\right)F(s) = 8e^{-9s} - 5s - 3$$

$$F(s) = e^{-9s}\left(\frac{1}{s-3} - \frac{1}{s+5}\right) - \frac{11}{4}\frac{1}{s+5} - \frac{9}{4}\frac{1}{s-3}$$

$$f(t) = u_9(t)\left(e^{3(t-9)} - e^{-5(t-9)}\right) - \frac{11}{4}e^{-5t} - \frac{9}{4}e^{3t}$$

The convolution theorem

Theorem

$$f * g = g * f \qquad \textit{commutative law}$$
$$f * (g_1 + g_2) = f * g_1 + f * g_2 \qquad \textit{distributive law}$$
$$(f * g) * h = f * (g * h) \qquad \textit{associative law}$$
$$f * 0 = 0 * f = 0$$

Theorem

If $F(s) = \mathcal{L}\{f(t)\}$ *and* $G(s) = \mathcal{L}\{g(t)\}$ *both exist for* $s > a \geq 0$, *then*

$$\mathcal{L}\{(f * g)(t)\} = F(s)G(s), s > a.$$

Exercise

Use the Laplace Transform to solve the following ODE:

$$\phi(t) + \int_0^t (t - \xi)\phi(\xi)d\xi = 1$$

Solution

$$\phi(t) + \int_0^t (t-\xi)\phi(\xi)d\xi = 1$$

The Laplace transform of a function $y(t)$ is defined as

$$Y(s) = \mathcal{L}\{y(t)\} = \int_0^\infty e^{-st} y(t)dt$$

Consequently, the convolution theorem for it is

$$\mathcal{L}\left\{\int_0^t f(t-\tau)g(\tau)d\tau\right\} = F(s)G(s)$$

Take the Laplace transform of both sides of the integral equation.

$$\mathcal{L}\left\{\phi(t) + \int_0^t (t-\xi)\phi(\xi)d\xi\right\} = \mathcal{L}\{1\}$$

Use the fact that the transform is a linear operator.

$$\mathcal{L}\{\phi(t)\} + \mathcal{L}\left\{\int_0^t (t-\xi)\phi(\xi)d\xi\right\} = \mathcal{L}\{1\}$$

solution

Apply the convolution theorem.

$$\mathcal{L}\{\phi(t)\} + \mathcal{L}\{t\}\mathcal{L}\{\phi(t)\} = \mathcal{L}\{1\}$$

Evaluate the Laplace transforms.

$$\Phi(s) + \left(\frac{1}{s^2}\right)\Phi(s) = \frac{1}{s}$$

Solve for $\Phi(s)$.

$$\Phi(s)\left(1 + \frac{1}{s^2}\right) = \frac{1}{s}$$

$$\Phi(s) = \frac{s}{s^2 + 1}$$

Thus, $\phi(t) = cos(t)$